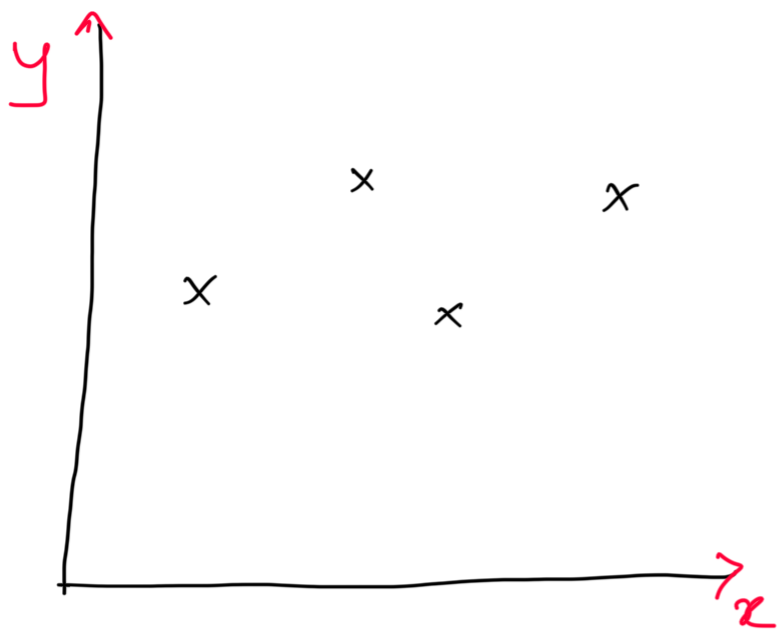
Gradient Descent



Suppose, We have a dataset, & on which we want to perform a linear regression which means we want to Find best Fit line.

This means we want toget to know the value of m & b

Consider $m = 78.35$

and we want to apply gradient descent to optimize value of b.

(For beginning it is easy to learn gradient descent by considering some value of m)

Later on we will take both to study the algorithm

step 1 - start with a random value of b

Consider $b = b$

For i in epoch:

$$b_{new} = b_{old} - \eta \times slope$$

η = learning rate = 0.01

Now,

To calculate slope

We know

$$Loss = L = \sum_{i=1}^{n} (y - \hat{y})^2$$

Loss function

$$\frac{dL}{db} = \frac{d}{db}\left[\sum (y - mx - b)^2\right]$$

$$= 2\left[\sum (y - mx - b)\right] \frac{d}{db}(y - mx - b)$$

$$\boxed{= -2\left[\sum (y - mx - b)\right]}$$

Consider $b = 0$

$$= -2\left[\sum (y - 78.35x - 0)\right]$$

$$\therefore \quad b_{new} = b_{old} - \eta \, \text{slope}_{b = b_{old}}$$